Unit 2 Assignment 4

Statistical Analysis of the Titanic Dataset — Group Report

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This report summarizes our statistical analysis of the Titanic dataset, loaded with `sns.load_dataset('titanic')`. All numbers come from the Jupyter Notebook.

Task 1 — Dataset Exploration

- **Size:** 891 rows and 15 columns.
- **Columns:** survived, pclass, sex, age, sibsp, parch, fare, embarked, class, who, adult_male, deck, embark_town, alive, alone.
- **Data types:** int (survived, pclass, sibsp, parch), float (age, fare), text/object (sex, embarked, who, embark_town, alive), category (class, deck), bool (adult_male, alone).
- **Missing values:** deck 688 (77%), age 177 (20%), embarked 2, embark_town 2. All other columns are complete.

Description: The dataset contains information about 891 passengers of the Titanic, which sank in 1912. For each passenger it records personal details (sex, age), travel details (ticket class, fare, port of embarkation, deck) and family on board. The column *survived* shows whether the passenger survived (1) or not (0). Some columns repeat the same information in another form, for example *class* and *pclass*, or *alive* and *survived*. The dataset is widely used to practice data analysis and to predict survival with Machine Learning.

Task 2 — Variable Classification

Variable	Description	Numerical / Categorical	Type
survived	0 = No, 1 = Yes	Categorical	Nominal
pclass	Ticket class 1, 2, 3	Categorical	Ordinal
sex	male / female	Categorical	Nominal
age	Age in years	Numerical	Continuous
sibsp	Siblings/spouses on board	Numerical	Discrete
parch	Parents/children on board	Numerical	Discrete
fare	Ticket price	Numerical	Continuous
embarked	Port: C, Q, S	Categorical	Nominal
class	First, Second, Third	Categorical	Ordinal
who	man / woman / child	Categorical	Nominal
adult_male	True / False	Categorical	Nominal
deck	Deck A to G	Categorical	Ordinal
embark_town	Cherbourg, Queenstown, Southampton	Categorical	Nominal
alive	yes / no	Categorical	Nominal
alone	True / False	Categorical	Nominal

Task 3 — Features and Target

1. **Target variable:** *survived*.
2. **Input features:** pclass, sex, age, sibsp, parch, fare, embarked (also who, adult_male, alone, deck). We exclude *alive*, because it is the same as the target, and *class* and *embark_town*, because they repeat pclass and embarked.
3. **Why the target is the dependent variable:** its value depends on the other variables. It is what we want to explain or predict.
4. **Why the others are independent variables:** they are the inputs the model uses to make the prediction. The model does not predict them.

Task 4 — Population and Sampling

	Size	Not survived (0)	Survived (1)
Population (all passengers)	891	61.62%	38.38%
Random sample (30%)	267	58.80%	41.20%
Stratified sample (30%, by survived)	268	61.57%	38.43%

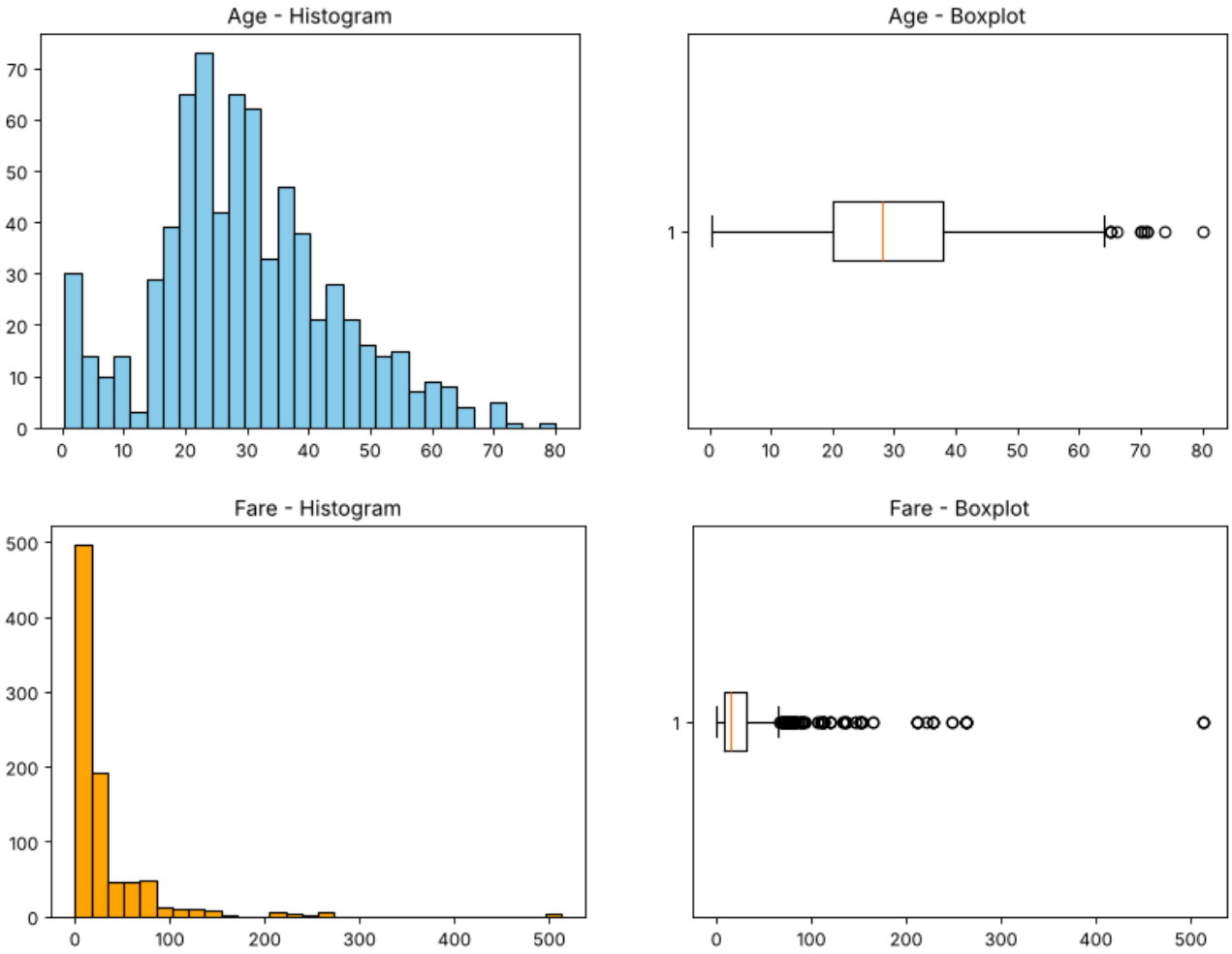
1. **Population:** all 891 passengers in the dataset.
2. **Sample:** a smaller group taken from the population, here 30% of the passengers.
3. **Sizes:** each sample is about 30% of the population (267 and 268 rows). The stratified sample has one extra row because 30% of each class is rounded separately.
4. **The stratified sample** better preserves the original class distribution: it is almost identical to the population, while the random sample has about 3% more survivors.
5. **Why it matters:** if a sample does not look like the real data, the model learns a wrong picture and makes biased predictions. A representative sample keeps training and testing fair and reliable, especially when one class is smaller than the other.

Task 5 — Descriptive Statistics

Variable	Mean	Median	Mode	Min	Max	Range	Variance	Std
age	29.70	28.00	24.00	0.42	80.00	79.58	211.02	14.53
fare	32.20	14.45	8.05	0.00	512.33	512.33	2469.44	49.69
sibsp	0.52	0.00	0.00	0.00	8.00	8.00	1.22	1.10

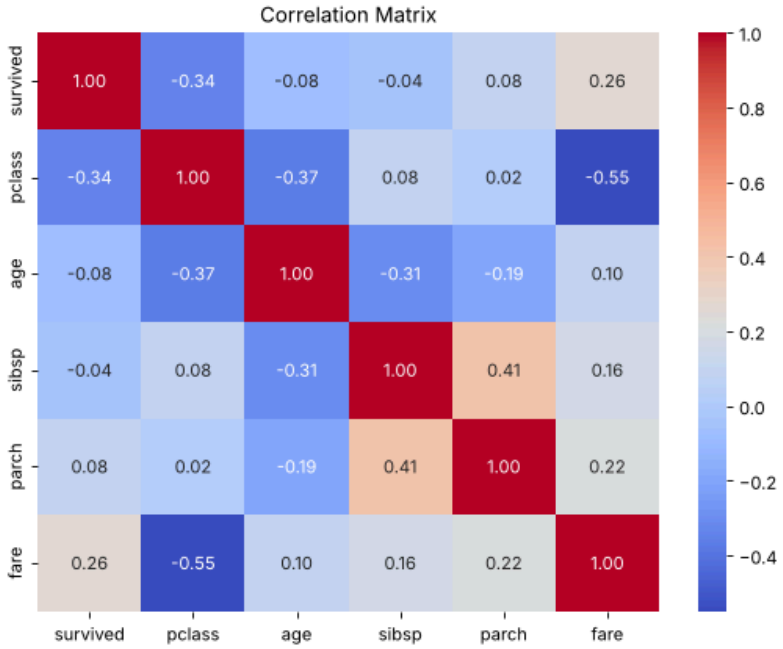
1. **Highest average:** fare (32.20).
2. **Greatest variability:** fare (std 49.69, variance 2469.44, range 512.33).
3. **Mean vs median:** close for age (29.70 vs 28.00); far apart for fare (32.20 vs 14.45); the mean is higher than the median for sibsp (0.52 vs 0).
4. **Distribution:** age is almost symmetric (slightly right-skewed). Fare and sibsp are right-skewed: most passengers paid low fares and had few relatives, but a few very large values pull the mean up.

Task 6 — Distribution and Outlier Analysis



1. **Shape:** age is approximately symmetric, slightly right-skewed. Fare is strongly right-skewed, with a long tail of expensive tickets.
2. **Outliers (IQR rule):** age has 11 (passengers older than about 65). Fare has 116 (fares above about 65.6, up to 512.33).
3. **The mean is more affected by outliers** than the median.
4. **From the plots:** in the fare boxplot many points lie far to the right of the box. These extreme values pull the mean (32.20) up, while the median (14.45), the line inside the box, stays in the middle of the data. Age has few outliers, so its mean and median stay close.

Task 7 — Correlation and Covariance Analysis



Covariance (selected values): variance of fare = 2469.44, variance of age = 211.02, cov(age, fare) = 73.85, cov(pclass, fare) = −22.83.

1. **Strongest positive correlation:** sibsp and parch (0.41), followed by fare and survived (0.26). The strongest relationship overall is negative: pclass and fare (−0.55).
2. **Little or no relationship:** sibsp and survived (−0.04), parch and pclass (0.02), age and survived (−0.08).
3. **Difference:** both show whether two variables move together. Covariance depends on units and scale, so it can be any number. Correlation is covariance divided by both standard deviations, so it is always between −1 and 1.
4. **Correlation is easier to interpret** because it has no units and a fixed scale, so strengths can be compared directly. Covariance values are hard to compare because large-scale variables like fare give large numbers even for weak relationships.

Task 8 — Statistical Insights

1. **Fare has the largest spread:** std 49.69 and range 512.33, much larger than any other variable.
2. **Fare contains the most outliers (116):** it is strongly right-skewed, while age has only 11 outliers.
3. **Related variables:** pclass and fare are strongly related (−0.55), because higher classes paid more. sibsp and parch are positively related (0.41).
4. **Fare has the highest average (32.20),** but because of skewness the median (14.45) better describes a typical passenger.
5. **Variables that need preprocessing:** deck is 77% missing, age is 20% missing, fare is skewed with outliers, and only 38% of passengers survived, so the classes are imbalanced.

Task 9 — Statistics for Machine Learning

Group	Survival rate
Female	74.2%
Male	18.9%
1st class	63.0%
2nd class	47.3%
3rd class	24.2%

1. **Most useful feature: sex.** 74% of women survived compared to 19% of men. Among numerical variables, pclass has the strongest correlation with survival (−0.34), followed by fare (0.26).
2. **Features to remove:** *alive*, because it is the target itself and would leak the answer. *class* and *embark_town*, because they duplicate pclass and embarked. *deck*, because 77% of it is missing.
3. **PCA:** in the covariance matrix, fare (variance 2469) and age (variance 211) have by far the largest variances, so without scaling they would dominate the first principal components. After standardizing, the correlation matrix shows that pclass–fare (−0.55) and sibsp–parch (0.41) share information, so these groups would contribute most.
4. **Preprocessing:** fill missing age with the median (28) and embarked with the mode ('S'); drop deck, alive, class and embark_town; encode categorical variables (sex → 0/1, one-hot encode embarked); reduce fare skewness with a log transform; scale numerical features, especially before PCA; and use a stratified train/test split, since only 38% survived.